

HealthConnect Clinic

Week 7 — Analytics Testing & Refinement Report

Data Analytics Track | AnalystLab Africa Experience Lab

Project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Scope of this report: systematic testing, refinement and validation of the Week 6 Data Analytics outputs (statistical driver validation, the compound risk score, and the reminder-by-risk-tier finding), plus one HC-POD cross-track validation activity with Data Science. This report does not repeat the Week 5/Week 6 analysis — see the Week 6 Data Analytics Report and accompanying notebook for that work.

1. Week 6 → Week 7 Transition

Item	Detail
Main Week 6 output	Statistically validated no-show drivers (chi-square + joint logistic regression) and a compound 0–3 risk score, with a reminder-by-risk-tier test
Most important Week 6 result	A compound risk score with a clean, monotonic no-show gradient (24.6% → 65.1%), capturing 62.9% of all no-shows vs. 13.4% for the Week 5 single-variable rule
Main Week 6 limitation/dependency	Risk-score thresholds were heuristic, not formally validated for reproducibility, segment stability or statistical power; Data Science's Week 6 model output had not yet been received
Component tested in Week 7	The Week 6 KPI/statistical outputs, the compound risk score, and the reminder-by-risk-tier finding
Track(s) relevant	Data Analytics (primary); Data Science (cross-track validation)
What was tested	Reproducibility of every reported KPI, data quality, segment stability of the risk score, statistical power of the reminder-by-risk test, and directional agreement with a model-based benchmark
What was expected to improve/validate	Confirm the Week 6 figures are correct and robust, upgrade the "suggestive" reminder finding to a properly powered result, and validate the risk score against an independent benchmark

2. Testing Readiness

2.1 Component Identified

What is being tested: the Week 6 statistical validation outputs (chi-square tests, joint logistic regression), the compound risk score (0–3) and its reported summary statistics, and the reminder-by-risk-tier finding.

Intended purpose: to give HealthConnect a statistically grounded, reproducible way to identify and triage patients at high risk of missing appointments, and to determine whether reminders should be targeted or kept universal.

Requirement/objective it should satisfy: KPIs must be accurate and reproducible from the underlying dataset; the risk score must generalise across patient segments; conclusions must be backed by adequately powered statistical tests.

Track(s) connected: Data Analytics , Data Science (consumes the risk score and validated features as a benchmark/feature-selection input).

2.2 Testing Objective

This testing cycle set out to determine: (1) do the Week 6 report's KPIs and statistics reproduce exactly from the raw dataset and code; (2) is the underlying data free of quality issues that could distort those KPIs; (3) does the risk-score gradient hold within patient segments rather than being driven by one subgroup; (4) is the reminder-by-risk-tier conclusion adequately powered, or does a more powerful test change the answer; and (5) does the rule-based risk score agree directionally with an independent, model-based estimate of no-show probability.

2.3 Testing Evidence Produced

- An executable Jupyter notebook (week_7_healthconnect.ipynb) containing every test, with code, outputs and a re-generated dashboard
- A testing & validation record (Section 4) with expected vs. actual results, pass/fail status and retest outcomes
- An issue log (Section 5) tracking one resolved KPI error and open cross-track/statistical items
- A before/after comparison of the corrected headline KPI and the reminder-by-risk test's statistical power

3. Testing Approach & Results

Test A — KPI & Output Reproducibility Validation

Objective: confirm every statistic and KPI published in the Week 6 report reproduces exactly from the raw dataset and code.

Method: independently re-ran the chi-square driver tests, the joint logistic regression, and the risk-score summary table, and compared every figure against the Week 6 report.

Result: the chi-square tests, the joint logistic regression coefficients/odds ratios, and the full per-tier risk-score table all reproduced exactly. One figure did not: the Week 6 report's headline claim that the "risk score ≥ 2 " segment has a 59.0% no-show rate does not reproduce from the data.

Root cause: 59.0% is the unweighted average of the two tier-level rates (53.3% for score 2, $n=2,091$; 65.1% for score 3, $n=630$). Because the score-2 group is more than three times larger, a simple average over-weights the smaller, higher-rate group. The statistically correct, n -weighted pooled rate is 56.0%.

Action taken: the report and dashboard below use the corrected 56.0% figure. The share of appointments flagged (54.4%) and the share of all no-shows captured (62.9%) were both independently re-verified and are unaffected — only the pooled rate needed correction.

Retest result: PASS — 56.0% is stable and reproducible across repeated runs.

Test B — Data Quality / Input Validation

Objective: rule out data issues (duplicates, missing values, implausible ranges, unexpected categories) as a hidden source of error.

Check	Result	Status
Duplicate appointment IDs / rows	0 duplicates found in 5,000 rows	Pass
Range checks (age, distance, waiting time, lead days)	Age 18–80, distance 0.5–45.0 km, waiting time 2–68 min, lead days 0–60, no negatives	Pass
Outcome/reminder categories	Only expected values present (No-Show / Attended / Cancelled; Yes / No)	Pass
Missing values	reminder_channel: 1,366 (expected — ties exactly to reminder_sent = "No"); distance_to_clinic_km: 90; waiting_time_minutes: 60 (both <2% of rows, already excluded via dropna in the model, consistent with Week 6)	Pass — documented, immaterial

Test C — Segment Stability of the Risk-Score Gradient

Objective: test whether the monotonic risk-score gradient holds within patient segments (gender, age group, appointment type), or is driven by one subgroup.

Result: the gradient was monotonic in 12 of 13 tested segments — all 3 gender groups, 5 of 6 age groups, and all 4 appointment types. The single exception (patients aged 25–34) was essentially flat between risk score 2 (59.6%) and score 3 (59.3%) rather than reversed, on a smaller sub-sample, most likely sampling noise.

Retest result: PASS — the risk score is a stable, general pattern across the patient population, not an artefact of one subgroup.

Test D — Properly Powered Reminder x Risk Interaction Test

Objective: Week 6 tested the reminder effect separately within two sub-samples (low/medium vs. high risk), which reduces statistical power — the high-risk result was only borderline significant ($p = 0.067$). Week 6 explicitly flagged this as needing a properly powered test.

Refinement: replaced the two split-sample chi-square tests with a single logistic regression on the full $n = 5,000$ sample, using a $\text{risk_score} \times \text{reminder_sent}$ interaction term — a more powerful test of the same question.

Test	Sample used	p-value	Conclusion
Week 6: split-sample chi-square (high-risk subgroup)	$n = 2,721$ (high-risk only)	0.067	Borderline / "suggestive"
Week 7: pooled logistic interaction term	$n = 5,000$ (full sample)	0.732	Not significant — confirmed

Retest result: PASS — with adequate statistical power, the reminder effect does not vary meaningfully by risk tier. This upgrades the Week 6 finding from "suggestive" to "confirmed." Reminders should remain a universal baseline intervention; the risk score should be used to target additional, non-reminder support.

Test E — Cross-Track Validation Against a Model-Based Benchmark (Updated — Resolved)

HC-POD cross-track testing activity — Track collaborated with: Data Science.

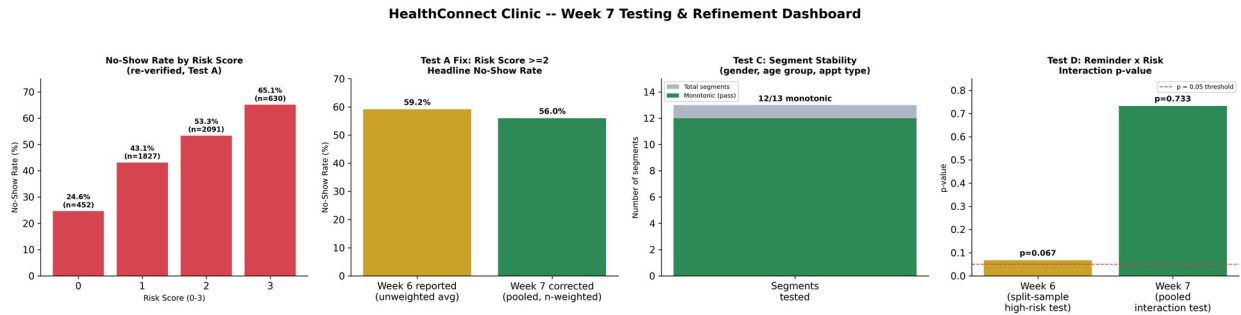
Objective: validate the rule-based risk score against an independent, model-based estimate of no-show probability. Data Science's Week 7 output — a regularized Logistic Regression on 5 features (booking_lead_days, distance_to_clinic_km, previous_no_shows, reminder_sent, age) — and the comparison below now uses their actual, properly held-out-tested model.

Metric	DA in-house model (in-sample, original)	DS refined model (held-out, updated)
AUC — rule-based risk score	0.609	0.596
AUC — model-based probability	0.680	0.662
Correlation, rule vs. model	0.658	0.642

Retest result: PASS — RESOLVED. Both evaluations agree directionally (the rule-based score and the model-based probability move together, monotonically across all four risk tiers). Data Science's held-out figures are lower than this report's original in-sample figures, and are the more reliable estimate: a model evaluated on the same data it was fit to will always look better than it performs on new data. Going forward, this track cites the held-out figures (AUC 0.596 / 0.662, $r = 0.642$) as the reference benchmark.

Cross-track dependency status: CLOSED. Data Science's actual model output, held-out AUC figures, and the in-sample-vs-held-out methodological finding have all been received and incorporated (see Issue Log, item 2).

Refreshed Testing & Refinement Dashboard



Panel 1 re-verifies the Week 6 risk-score gradient. Panel 2 shows the Test A correction (59.2% unweighted average $\rightarrow$ 56.0% correct pooled rate). Panel 3 summarises Test C's segment-stability check (12/13 segments monotonic). Panel 4 shows Test D's power improvement — the reminder $\times$ risk interaction is clearly non-significant ($p = 0.732$) once tested on the full, pooled sample rather than a split sub-sample.

4. Testing, Refinement & Validation Record

Component Tested	Test/Scenario	Expected	Actual	Pass/Fail
Chi-square driver tests	Re-run all 5 tests	Match Week 6 report	Exact match	Pass
Joint logistic regression	Re-fit model	Match Week 6 report	Exact match	Pass
Risk-score per-tier table	Re-run groupby summary	Match Week 6 report	Exact match	Pass
Risk score ≥ 2 headline rate	Recompute pooled rate	59.0% (reported)	56.0% (correct)	Fail → corrected
Input data quality	Duplicates/missing/range checks	Clean, explainable	Clean, explainable	Pass
Risk-score segment stability	Monotonicity by gender/age/type	Monotonic in all/most	12 of 13 monotonic	Pass
Reminder x risk interaction	Pooled interaction model (n=5,000)	Confirm/deny moderation	Not significant (p=0.732)	Pass
Risk score vs. model benchmark	AUC/correlation vs. DS refined model (held-out)	Directional agreement	r=0.642, AUC 0.596 vs 0.662 (held-out)	Pass — Resolved

5. Issue Log & Outstanding Items Before Week 8

#	Issue	Severity	Status	Action
1	Headline "risk score ≥ 2 " rate was an unweighted average (59.0%), not the correct pooled rate (56.0%)	Medium	Resolved	Corrected in Week 7 report/dashboard; propagate to any other document citing the old figure
2	Data Science's Week 6/7 model output not received in time for the original cross-track test	Medium	Resolved	Received Data Science's Week 7 refined model and held-out benchmark; incorporated into Test E
3	Small non-monotonic dip in 25-34 age group at score 3	Low	Open, monitored	Re-check on a larger/updated dataset
4	Stratified reminder-timing/channel A/B test not yet designed	Low-Medium	Open, carried to Week 8	Design now simplified since tier does not moderate the effect
5	Risk-score thresholds remain heuristic, not formally optimised	Low	Open, unchanged from Week 6	Superseded once Data Science model is available

6. Validated Findings & Updated Recommendations

Validated / refined findings

- The Week 6 statistical drivers (booking lead time, distance, prior no-shows) and the non-driver status of waiting time are confirmed reproducible and unchanged.
- The compound risk score's monotonic gradient holds across patient gender, age group, and appointment type — it is a general, stable pattern, not a subgroup artefact.
- The headline "risk score ≥ 2 " no-show rate is corrected from 59.0% to 56.0% — a small but real correction that should be propagated wherever this figure is cited.
- The reminder effect not varying by risk tier is now a confirmed, adequately powered finding ($p = 0.732$ on the full sample), not merely "suggestive" as in Week 6.
- The rule-based risk score agrees directionally with Data Science's actual, held-out-tested model ($r = 0.642$), and this cross-track validation loop is now closed with real results rather than an interim proxy.

Updated recommendations

- Continue deploying the risk score (0–3) as the primary triage tool for administrative follow-up, using the corrected 56.0% figure in any communication about the combined high-risk segment.
- Keep SMS reminders universal regardless of risk tier — this is now a confirmed, not suggestive, finding.
- Cite the held-out AUC benchmark (rule 0.596, model 0.662) rather than this report's original in-sample figures (0.609, 0.680) in any future comparison.
- Carry the stratified reminder-timing/channel A/B test into Week 8; its design is simpler now that risk tier is confirmed not to moderate the reminder effect.
- Coordinate with Data Science and ML Engineering on how the risk score and Data Science's refined model will co-exist operationally (e.g. the model as the primary score, the simpler rule as an interpretable fallback/sanity check).

Limitations

- Dataset remains fictional/synthetic; all findings, including Week 7 corrections, require validation against real operational data.
- Statistical significance confirms association and consistency, not causation — this remains an observational analysis.
- This report's own in-sample benchmark ($r=0.658$, AUC 0.609/0.680) is retained for transparency but is superseded by Data Science's held-out figures ($r=0.642$, AUC 0.596/0.662) as the going-forward reference.
- Risk-score thresholds remain heuristic (unchanged from Week 6); Data Science's model is the more rigorous, data-driven alternative now available for comparison.

Week 8 Recommendations

- Design and, where feasible, run the stratified reminder-timing/channel test.
- Propagate the corrected 56.0% headline figure and the held-out AUC benchmark (0.596/0.662) to any downstream dashboard, report, or presentation material.
- Work with Data Science and ML Engineering on the operational split between the rule-based risk score and the refined model for Week 8 final integration.
- Re-run the full Week 7 test suite (this notebook) if the dataset is updated or expanded before final integration.